

Proximal Balancing

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We study causal adjustment with observed pretreatment covariates X , one pretreatment proxy W , and an unobserved confounder. In the canonical regime, W supervises a finite representation $Z = \phi(X)$ but is not included in the final adjustment variable. We establish an exact bias decomposition and show that, under a uniform outcome-relevant stability condition, the observable residual imbalance $D(\phi)$ yields the sensitivity certificate $|\tau_\phi - \tau| \leq \Gamma D(\phi)$. Exact balance identifies the average treatment effect under the stated proxy conditions, while a finite-class learning analysis controls the certificate achieved by empirical representation selection. We separately study a proxy-augmented audit-view construction that prevents literal self-conditioning; its point-identification result requires additional multi-view structure and remains a conditional extension. Accordingly, we do not claim that one generic proxy alone point-identifies the average treatment effect.

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1. Introduction

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given X , conditioning on $e(X) = P(A = 1 | X)$ balances the distribution of X across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. Our setting keeps a latent confounder and asks what an observed proxy can certify about the bias remaining after a learned reduction of X .

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). Our construction shares the use of a learned low-dimensional representation and a distributional balance penalty. The difference is the role of the proxy: W is not an outcome predictor in the canonical method, but an outcome-free supervision signal for choosing $Z = \phi(X)$ when ignorability given X may fail.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by the treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024); semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). Our canonical result has a different target: it uses one pretreatment proxy to bound the bias of adjustment for $Z = \phi(X)$, and it treats an outcome bridge as a structural witness for a sensitivity budget rather than as an estimated nuisance function.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population ATE target, from our latent-confounder proxy interpretation, and from our use of an observable balance discrepancy. The comparison is nevertheless central: it makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge, an adjustment score, or an independent audit view.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero for identical distributions when its kernel is characteristic (Gretton et al., 2012). We use conditional treatment-arm MMDs of (X, W) as observable magnitudes, not kernel-test p -values. The new theoretical object is the product of this discrepancy and an outcome-relevant stability constant, which yields a bias certificate and a representation-selection guarantee. The individual ingredients have close precedents in balancing, representation learning,

kernel testing, and proximal inference; the manuscript therefore makes no priority claim before a systematic literature audit, and instead states its contribution through the exact assumptions, estimand, and guarantee proved below.

2. Problem Setting

2.1. Preliminaries

Structural causal models. A structural causal model (SCM) consists of endogenous variables, exogenous disturbances, and structural assignments. Writing the endogenous variables as $V_1, \dots, V_d$, an acyclic SCM has equations

$$V_j = f_j(\text{pa}_j, E_j), \quad j = 1, \dots, d, \quad (1)$$

where pa_j contains the direct causes of V_j and E_j is an exogenous disturbance. The intervention $\text{do}(A = a)$ replaces the structural equation for A by the constant a while leaving the other equations unchanged. The resulting outcome is the potential outcome $Y(a)$. An average causal effect is *identified* when its value is uniquely determined by the observed-data distribution for every SCM in the stated model class (Pearl, 2009).

Proximal causal inference. Ordinary adjustment seeks observed covariates that block all non-causal paths from treatment to outcome. Proximal causal inference instead allows an unobserved confounder and uses proxy measurements of it. Standard proximal formulations assign two different proxy roles and posit a bridge equation whose solution translates an observable conditional moment into the target causal functional (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Completeness or injectivity prevents distinct latent laws from producing the same observable proxy law. These conditions do not follow from the mere existence of a correlated proxy. Our method retains this distinction: injectivity is used for exact interpretation, while quantitative stability is used when balance is only approximate.

2.2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (2)$$

where X is a vector of observed pretreatment covariates, W is one observed pretreatment proxy, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. The variable U denotes an unobserved pretreatment cause that may affect both A and Y . The target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (3)$$

We do not assume that $A \perp\!\!\!\perp U \mid X$.

The canonical method fixes an integer $M \geq 1$, writes $[M] \triangleq \{1, \dots, M\}$, and prespecifies a class Φ

of measurable maps from the support of X to $[M]$. For $\phi \in \Phi$, the adjustment representation is

$$Z \triangleq \phi(X). \quad (4)$$

This is a construction, not an assumption. The proxy W supervises the choice of ϕ through an outcome-free balance criterion, but W is not included in the canonical adjustment variable Z .

For a fixed ϕ , define the population contrast obtained by adjusting for Z as

$$\tau_\phi \triangleq \sum_{z=1}^M P(Z = z) [\mathbb{E}(Y \mid A = 1, Z = z) - \mathbb{E}(Y \mid A = 0, Z = z)]. \quad (5)$$

The central questions are whether observable balance can imply $\tau_\phi = \tau$, how the difference $\tau_\phi - \tau$ can be bounded when balance is imperfect, and whether a representation minimizing an empirical balance criterion inherits a finite-sample causal guarantee. All additional notation is introduced only when the corresponding result requires it.

3. Identification Theory

This section develops one population argument. We first express the bias of adjustment for $Z = \phi(X)$ as a functional of the latent treated-versus-control contrast. We then show how the observable imbalance of (X, W) controls that functional under a declared stability budget. Exact balance is treated as an endpoint to be certified, not as a property that every representation class can attain.

Assumption 1 (Causal and proxy structure). *For each $a \in \{0, 1\}$, consistency and integrability hold,*

$$Y = Y(A), \quad \mathbb{E}|Y(a)| < \infty, \quad (6)$$

and treatment is conditionally exchangeable given the observed covariates and latent confounder,

$$Y(a) \perp\!\!\!\perp A \mid (X, U). \quad (7)$$

The proxy is pretreatment and its measurement law is invariant across treatment arms:

$$W(a) = W, \quad W \perp\!\!\!\perp A \mid (X, U). \quad (8)$$

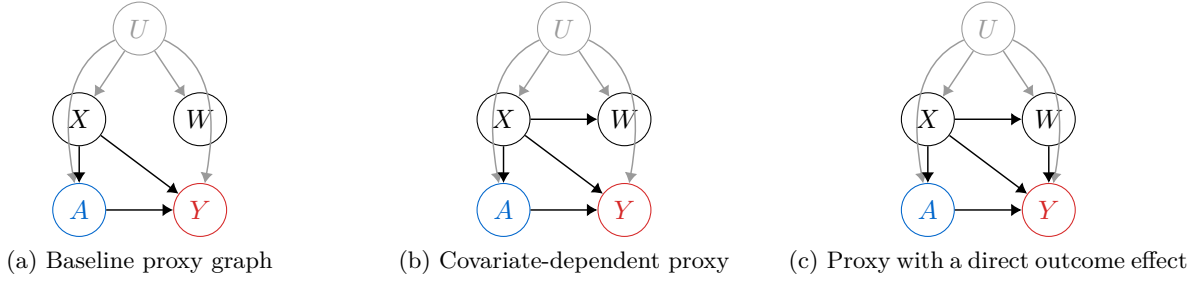


Figure 1: Three directed acyclic graphs compatible with Assumption 1. The panels are compatible examples under a Markovian structural causal model with mutually independent exogenous disturbances and no unshown common causes. Panel (b) allows X to affect the proxy. Panel (c) shows that Assumption 1 permits a pretreatment proxy to directly affect Y and does not impose an outcome exclusion restriction. The graphs do not encode the consistency or integrability clauses of Assumption 1, and they do not characterize all models allowed by the assumption.

The next assumption makes both the adjustment contrast and the observable balance criterion well defined for every retained stratum.

Assumption 2 (Stratum and kernel regularity). *There are constants $p_{\min} > 0$ and $\eta \in (0, 1/2)$ such that, for every retained $\phi \in \Phi$ and stratum z ,*

$$p_z \triangleq P(Z = z) \geq p_{\min}, \quad \eta \leq \pi_z \triangleq P(A = 1 \mid Z = z) \leq 1 - \eta. \quad (9)$$

The kernel k on the joint space of (X, W) is bounded and characteristic, and all kernel mean embeddings used below exist.

Definition 1 (Latent arm contrast). *For a fixed representation and stratum z , let*

$$Q_{a,z} \triangleq \mathcal{L}(X, U \mid A = a, Z = z), \quad \nu_{\phi,z} \triangleq Q_{1,z} - Q_{0,z}. \quad (10)$$

When it is first needed, define the latent outcome regression by

$$\mu_a(x, u) \triangleq \mathbb{E}\{Y(a) \mid X = x, U = u\}. \quad (11)$$

Theorem 1 (Exact bias decomposition). *Under the consistency, integrability, and latent-exchangeability parts of Assumption 1, together with the mass and overlap conditions in Assumption 2, every fixed $\phi \in \Phi$ satisfies*

$$\tau_\phi - \tau = \sum_{z=1}^M p_z \int \{(1 - \pi_z)\mu_1(x, u) + \pi_z\mu_0(x, u)\} d\nu_{\phi,z}(x, u). \quad (12)$$

Theorem 1 isolates the quantity that a proxy must control. Recovering every feature of the latent distribution is sufficient, but the bias depends only on the displayed outcome-relevant functionals.

Definition 2 (Observable joint discrepancy). *Let $\mathcal{H}_k$ denote the RKHS induced by k . For a probability law P on (X, W) , define its kernel mean by*

$$m_k(P) \triangleq \int k(v, \cdot) dP(v), \quad (13)$$

and define

$$\text{MMD}_k(P, Q) \triangleq \|m_k(P) - m_k(Q)\|_{\mathcal{H}_k}. \quad (14)$$

For each treatment arm and stratum, let

$$R_{a,z} \triangleq \mathcal{L}(X, W \mid A = a, Z = z). \quad (15)$$

The discrepancy in stratum z and its population aggregate are

$$\Delta_z(\phi) \triangleq \text{MMD}_k(R_{1,z}, R_{0,z}), \quad (16)$$

$$D(\phi) \triangleq \sum_{z:p_z>0} p_z \Delta_z(\phi). \quad (17)$$

Proposition 1 (Observable meaning of zero discrepancy). *Under Assumption 2,*

$$D(\phi) = 0 \iff (X, W) \perp\!\!\!\perp A \mid Z. \quad (18)$$

Proposition 1 is an equality of observed conditional laws. By itself, it does not eliminate hidden confounding. The next definitions state the additional link used by the approximate theory.

Definition 3 (Proxy transport and outcome functionals). *Assumption 1 gives the treatment-invariant proxy channel*

$$K_x(B \mid u) \triangleq P(W \in B \mid X = x, U = u). \quad (19)$$

For a finite signed measure ν on $\mathcal{X} \times \mathcal{U}$, define its image on $\mathcal{X} \times \mathcal{W}$ by

$$(\mathcal{K}\nu)(C) \triangleq \int \int 1\{(x, w) \in C\} K_x(dw \mid u) \nu(dx, du). \quad (20)$$

Before observing the data, fix a linear space $\mathcal{C}_{\phi,z}$ of integrable, total-mass-zero contrasts that contains every $Q_{1,z} - Q_{0,z}$ allowed by the scientific model. For $\nu \in \mathcal{C}_{\phi,z}$, define

$$J_{\phi,z}\nu \triangleq \int k(v, \cdot) d(\mathcal{K}\nu)(v) \in \mathcal{H}_k, \quad (21)$$

$$\ell_{a,\phi,z}(\nu) \triangleq \int \mu_a(x, u) d\nu(x, u). \quad (22)$$

Assumption 3 (Uniform outcome-relevant stability). *For each a, ϕ , and retained z , there*

is a finite $\Gamma_{a,\phi,z}$ such that

$$|\ell_{a,\phi,z}(\nu)| \leq \Gamma_{a,\phi,z} \|J_{\phi,z}\nu\|_{\mathcal{H}_k}, \quad \nu \in \mathcal{C}_{\phi,z}. \quad (23)$$

For representation selection with the unweighted criterion $D(\phi)$, these constants share a prespecified finite budget,

$$\sup_{\phi \in \Phi} \max_{z: p_z > 0} \max_{a \in \{0,1\}} \Gamma_{a,\phi,z} \leq \Gamma < \infty. \quad (24)$$

The budget Γ is a sensitivity parameter unless external information calibrates it. Injectivity alone does not supply a quantitative common budget, and Γ is not generally identified from one proxy.

Lemma 1 (Channel identity). *Under the proxy-invariance part of Assumption 1,*

$$R_{a,z} = \mathcal{K}Q_{a,z}, \quad \|J_{\phi,z}\nu_{\phi,z}\|_{\mathcal{H}_k} = \Delta_z(\phi). \quad (25)$$

The channel identity puts the latent contrast in Theorem 1 and the observed discrepancy in Definition 2 on the same scale. Stability then converts this equality into the main population guarantee.

Theorem 2 (Population proxy-balancing certificate). *Under Assumptions 1, 2, and 3, every fixed $\phi \in \Phi$ obeys*

$$|\tau_\phi - \tau| \leq \sum_{z=1}^M p_z \{(1 - \pi_z)\Gamma_{1,\phi,z} + \pi_z\Gamma_{0,\phi,z}\} \Delta_z(\phi). \quad (26)$$

Under the common budget Γ , this becomes

$$|\tau_\phi - \tau| \leq \Gamma D(\phi). \quad (27)$$

Proof. Apply Theorem 1, the triangle inequality, Assumption 3, and Lemma 1. The weights on the two treatment-specific constants sum to one within each stratum, which gives the common-budget form. $\square$

The central theorem controls the certified worst-case bias within the declared model class. It does not claim that decreasing $D(\phi)$ must decrease the unknown realized bias for every data-generating process.

Proposition 2 (Minimal stability and bounded factorization). *For fixed (a, ϕ, z) , define*

$$\Gamma_{a,\phi,z}^* \triangleq \sup_{\substack{\nu \in \mathcal{C}_{\phi,z} \\ \|J_{\phi,z}\nu\|_{\mathcal{H}_k} > 0}} \frac{|\ell_{a,\phi,z}(\nu)|}{\|J_{\phi,z}\nu\|_{\mathcal{H}_k}}. \quad (28)$$

Set $\Gamma_{a,\phi,z}^ = \infty$ if $J_{\phi,z}\nu = 0$ but $\ell_{a,\phi,z}(\nu) \neq 0$ for an admissible ν , and set an empty supremum*

to zero when $\ell_{a,\phi,z}$ vanishes on the kernel. Then $\Gamma_{a,\phi,z}^* < \infty$ if and only if there is a bounded linear functional $\tilde{\ell}_{a,\phi,z}$ on $J_{\phi,z}(\mathcal{C}_{\phi,z})$ such that

$$\ell_{a,\phi,z} = \tilde{\ell}_{a,\phi,z} \circ J_{\phi,z}. \quad (29)$$

This functional extends uniquely to the closure of the range and has a minimum-norm Riesz representer

$$g_{a,\phi,z}^* \in \overline{J_{\phi,z}(\mathcal{C}_{\phi,z})} \subseteq \mathcal{H}_k \quad (30)$$

satisfying

$$\ell_{a,\phi,z}(\nu) = \langle g_{a,\phi,z}^*, J_{\phi,z}\nu \rangle_{\mathcal{H}_k}, \quad \|g_{a,\phi,z}^*\|_{\mathcal{H}_k} = \Gamma_{a,\phi,z}^*. \quad (31)$$

This factorization states the precise price of the certificate. Only outcome-relevant latent contrasts must pass continuously through the observable proxy channel.

Proposition 3 (Global RKHS bridge witness). *For $g \in \mathcal{H}_k$, define*

$$(Tg)(x, u) \triangleq \int g(x, w) K_x(dw \mid u). \quad (32)$$

If, for each $a \in \{0, 1\}$, a single $g_a \in \mathcal{H}_k$ satisfies $Tg_a = \mu_a$ throughout the model class and $\|g_a\|_{\mathcal{H}_k} \leq \Gamma$, then Assumption 3 holds uniformly over all $\phi \in \Phi$ and retained strata.

The bridge in Proposition 3 is a structural witness, not an estimated object in the proposed method. It provides a common valid budget, but it need not provide the tight constant for any particular representation.

Corollary 1 (Exact identification through stability). *Under the assumptions of Theorem 2, if $D(\phi) = 0$, then $\tau_\phi = \tau$. Hence the ATE is point identified by the observed Z -adjustment functional defined in Section 2.2.*

Exact identification can also be interpreted through full latent balance. That interpretation uses the following stronger qualitative condition, which is deliberately introduced only here.

Assumption 4 (Conditional mixture injectivity). *For almost every x , the map*

$$\lambda \longmapsto \int K_x(\cdot \mid u) \lambda(du) \quad (33)$$

is injective over the admissible conditional laws of U given $X = x$. The admissible class contains both $\mathcal{L}(U \mid X = x, A = 0)$ and $\mathcal{L}(U \mid X = x, A = 1)$ wherever the argument is applied.

Theorem 3 (Latent interpretation of exact balance). *Under Assumptions 1, 2, and 4, if*

$D(\phi) = 0$, then

$$(X, U) \perp\!\!\!\perp A \mid Z, \quad Y(a) \perp\!\!\!\perp A \mid Z. \quad (34)$$

Consequently, $\tau_\phi = \tau$.

Proof. Zero discrepancy gives $(X, W) \perp\!\!\!\perp A \mid Z$. Because $Z = \phi(X)$, this implies both $X \perp\!\!\!\perp A \mid Z$ and $W \perp\!\!\!\perp A \mid X$. Proxy-channel invariance and Assumption 4 then give $U \perp\!\!\!\perp A \mid X$. Combining the last two independences yields $(X, U) \perp\!\!\!\perp A \mid Z$, after which latent exchangeability and consistency give the result. $\square$

Corollary 1 and Theorem 3 are two valid routes from an attained exact endpoint. Exact observed balance alone supplies neither route unless the corresponding stability or injectivity condition is imposed.

Theorem 4 (Finite- M attainability). *Let*

$$\Phi_M \triangleq \{\phi : \mathcal{X} \rightarrow [M] : \phi \text{ is measurable}\}, \quad (35)$$

let $e(X) \triangleq P(A = 1 \mid X)$, and assume primitive strict overlap $0 < e(X) < 1$ almost surely. Maps with empty or light strata are allowed, and $D(\phi)$ sums only over positive-probability strata. Assume that k is bounded and characteristic and that the required kernel mean embeddings exist. Then there exists $\phi \in \Phi_M$ with $D(\phi) = 0$ if and only if

$$W \perp\!\!\!\perp A \mid X \quad \text{and} \quad e(X) \text{ takes at most } M \text{ values almost surely.} \quad (36)$$

Theorem 4 separates certification from existence. Exact balance is a strong endpoint, while finite-stratum exact attainment is confined to a narrow boundary.

Proposition 4 (Restricted order preservation). *Fix a bounded characteristic kernel k_W on $\mathcal{W}$ and a bounded kernel k_X on $\mathcal{X}$, write $\mathcal{H}_{k_W}$ and $\mathcal{H}_{k_X}$ for their induced RKHSs, and define*

$$D_W(\phi) \triangleq \sum_{z=1}^M p_z \text{MMD}_{k_W} \{\mathcal{L}(W \mid A = 1, Z = z), \mathcal{L}(W \mid A = 0, Z = z)\}. \quad (37)$$

Assume all five conditions below.

(P1) $U \in \{0, 1\}$.

(P2) The proxy channel is x -homogeneous, $K_x(\cdot \mid u) = K(\cdot \mid u)$, and

$$u_W \triangleq \int k_W(w, \cdot) d\{K(\cdot \mid 1) - K(\cdot \mid 0)\}(w) \quad (38)$$

satisfies $\|u_W\|_{\mathcal{H}_{k_W}} > 0$.

(P3) The outcome regression has the additive homogeneous form $\mu_a(x, u) = f(x) + \tau a + \beta u$, where $f \in \mathcal{H}_{k_X}$.

(P4) Every compared representation has within-stratum X balance: $\mathcal{L}(X \mid A = 1, Z = z) = \mathcal{L}(X \mid A = 0, Z = z)$.

(P5) Every compared representation has a common direction of latent selection:

$$\mathbb{E}(U \mid A = 1, Z = z) - \mathbb{E}(U \mid A = 0, Z = z) \geq 0. \quad (39)$$

Under Assumption 1 and the overlap part of Assumption 2,

$$|\tau_\phi - \tau| = \kappa D_W(\phi), \quad \kappa \triangleq \frac{|\beta|}{\|u_W\|_{\mathcal{H}_{k_W}}}. \quad (40)$$

When $\beta \neq 0$, D_W and realized absolute bias therefore give the same strict ordering among representations satisfying P1–P5.

Proposition 4 is a restricted alignment result. Without its five conditions, Theorem 2 remains a worst-case certificate and does not generally order realized biases.

Remark 1 (Independent audit-view extension). For a high-dimensional proxy, one may write

$$W = (W^{\text{rep}}, W^{\text{audit}}), \quad R = \rho(X, W^{\text{rep}}), \quad (41)$$

and audit $(X, W^{\text{audit}}) \perp\!\!\!\perp A \mid R$. This construction prevents literal copying of the audited coordinate. Exact point identification still requires additional multi-view conditional independence and an outcome-relevant null-space or injectivity condition for the audit channel. Its exact theorem and approximate stability certificate remain conditional, open extensions and are not claims of the canonical $Z = \phi(X)$ theory above.

4. Estimating Causal Effects

The population certificate becomes operational in three steps. An outcome-free learner chooses a representation, independent data evaluate its observed discrepancy, and a finite-stratum adjustment estimator targets τ_ϕ . This section first analyzes finite representation classes, then states the fixed-representation semiparametric result, and finally combines the pieces in an honest finite-sample certificate.

Definition 4 (Empirical discrepancy and approximate ERM). For a candidate ϕ and cell (a, z) , let $n_{a,z}(\phi)$ be the cell count, let $\hat{p}_z(\phi)$ be the empirical stratum proportion, and let $\hat{\mu}_{a,z}^k(\phi)$ be the empirical kernel mean embedding of (X, W) in that cell. Define

$$\hat{\Delta}_z(\phi) \triangleq \|\hat{\mu}_{1,z}^k(\phi) - \hat{\mu}_{0,z}^k(\phi)\|_{\mathcal{H}_k}, \quad (42)$$

$$\hat{D}(\phi) \triangleq \sum_{z=1}^M \hat{p}_z(\phi) \hat{\Delta}_z(\phi). \quad (43)$$

An approximate empirical risk minimizer $\hat{\phi}$ satisfies

$$\hat{D}(\hat{\phi}) \leq \inf_{\phi \in \Phi} \hat{D}(\phi) + \varepsilon_{\text{opt}}, \quad \varepsilon_{\text{opt}} \geq 0. \quad (44)$$

Neither Y nor U enters this objective. Minimum stratum mass and representation-level overlap are enforced as admissibility requirements.

Theorem 5 (Finite-class uniform deviation). *Suppose Φ is finite, Assumption 2 holds uniformly over Φ , and*

$$\sup_v k(v, v) \leq \kappa^2. \quad (45)$$

For $\delta \in (0, 1)$, suppose

$$n \geq \frac{8}{p_{\min}\eta} \log \frac{8M|\Phi|}{\delta}. \quad (46)$$

Define

$$e_n(\delta) \triangleq \left(2\kappa + \kappa \sqrt{2 \log \frac{4M|\Phi|}{\delta}} \right) \sqrt{\frac{2}{np_{\min}\eta}}, \quad (47)$$

$$r_n(\delta) \triangleq 2e_n(\delta) + 2\kappa \sqrt{\frac{2\{M \log 2 + \log(4|\Phi|/\delta)\}}{n}}. \quad (48)$$

Then, with probability at least $1 - \delta$,

$$\sup_{\phi \in \Phi} |\hat{D}(\phi) - D(\phi)| \leq r_n(\delta). \quad (49)$$

Theorem 5 combines a cell-count event, bounded-kernel mean-embedding concentration, and a multinomial bound for the stratum weights. Its constants apply to an enumerable class and do not constitute a neural-network complexity bound.

Corollary 2 (Representation-selection oracle inequality). *Under Theorem 5, Definition 4, and the common stability budget in Assumption 3, with probability at least $1 - \delta$,*

$$D(\hat{\phi}) \leq \inf_{\phi \in \Phi} D(\phi) + 2r_n(\delta) + \varepsilon_{\text{opt}}, \quad (50)$$

$$|\tau_{\hat{\phi}} - \tau| \leq \Gamma \{ \hat{D}(\hat{\phi}) + r_n(\delta) \}, \quad (51)$$

$$|\tau_{\hat{\phi}} - \tau| \leq \Gamma \left\{ \inf_{\phi \in \Phi} D(\phi) + 2r_n(\delta) + \varepsilon_{\text{opt}} \right\}. \quad (52)$$

Remark 2 (Neural optimization is not the finite-class theorem). *The current neural implementation uses Adam and differentiable soft assignments before deploying or evaluating a hard representation. Optimization error may be recorded as ε_{opt} , but the present theory*

does not supply a neural-class analogue of $r_n(\delta)$ or a uniform soft-to-hard error bound. The training surrogate and the held-out hard discrepancy must therefore be reported separately.

The next construction separates representation choice from causal estimation. This separation lets the evaluation analysis condition on the learned map and treat it as fixed.

Definition 5 (Honest sample-splitting procedure). *Partition the observations into an independent representation-training sample and an evaluation sample. Learn $\hat{\phi}$ using only the training sample. On the evaluation sample, set $Z = \hat{\phi}(X)$, check empirical retained-cell mass and overlap diagnostics, compute a hard-stratum discrepancy $\hat{D}_{\text{ev}}(\hat{\phi})$, and estimate the adjusted contrast. Repeating this construction across outer folds may use every observation, but each fold is analyzed conditionally on the data used to learn its representation.*

Definition 6 (Fixed- Z AIPW estimator). *For a fixed finite representation, define*

$$e_Z(z) \triangleq P(A = 1 \mid Z = z), \quad (53)$$

$$m_a(z) \triangleq \mathbb{E}(Y \mid A = a, Z = z). \quad (54)$$

Given estimates $\hat{e}_Z$ and $\hat{m}_a$, the augmented inverse-probability weighted estimator is

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}_1(Z_i) - \hat{m}_0(Z_i) + \frac{A_i \{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}_Z(Z_i)} \right. \quad (55)$$

$$\left. - \frac{(1 - A_i) \{Y_i - \hat{m}_0(Z_i)\}}{1 - \hat{e}_Z(Z_i)} \right]. \quad (56)$$

Proposition 5 (Empirical-cell plug-in equals AIPW). *Suppose every retained empirical cell (a, z) is nonempty. Let $\hat{p}_z$ be the empirical stratum proportion and $\bar{Y}_{a,z}$ the empirical outcome mean in cell (a, z) . If $\hat{e}_Z$ is the empirical cell treatment proportion and $\hat{m}_a(z) = \bar{Y}_{a,z}$, then*

$$\hat{\tau}_{\text{AIPW}} = \sum_{z=1}^M \hat{p}_z \{\bar{Y}_{1,z} - \bar{Y}_{0,z}\}. \quad (57)$$

The equality follows because each inverse-probability residual term sums to zero within its empirical treatment-by-stratum cell. It is an algebraic fact for finite cells, not a claim about arbitrary machine-learned nuisances.

Theorem 6 (Fixed finite-representation efficiency). *Fix ϕ before observing the estimation sample, suppose $Z = \phi(X)$ is finite, representation-level overlap holds, and $\mathbb{E}(Y^2) < \infty$. The efficient influence function of τ_ϕ in the nonparametric model for (Z, A, Y) is*

$$\psi_\phi(O) = \frac{A\{Y - m_1(Z)\}}{e_Z(Z)} - \frac{(1 - A)\{Y - m_0(Z)\}}{1 - e_Z(Z)} + m_1(Z) - m_0(Z) - \tau_\phi. \quad (58)$$

The empirical-cell plug-in estimator in Proposition 5 satisfies

$$\sqrt{n}(\hat{\tau}_\phi - \tau_\phi) \rightsquigarrow \mathcal{N}\left(0, \mathbb{E}[\psi_\phi(O)^2]\right) \quad (59)$$

and is asymptotically efficient for this fixed- ϕ coarsened-data model.

The preceding theorem does not establish same-sample inference after adaptive representation selection. Honest splitting instead permits a conditional fixed-representation finite-sample statement.

Theorem 7 (End-to-end finite-sample sensitivity certificate). *Follow Definition 5. Assume that, for almost every realization of the training sample, the selected $\hat{\phi}$ satisfies the population mass and overlap conditions in Assumption 2 and its stability constants are bounded by Γ . Condition on such a training-sample realization so that $\hat{\phi}$ is fixed. Let the evaluation sample have size n_{ev} , assume $|Y| \leq B$, and let $\delta, \delta' \in (0, 1)$. Require*

$$n_{\text{ev}} \geq \max \left\{ \frac{8}{p_{\min}\eta} \log \frac{8M}{\delta}, \frac{8}{p_{\min}\eta} \log \frac{16M}{\delta'} \right\}. \quad (60)$$

Let $r_{n_{\text{ev}}}^{(1)}(\delta)$ be the radius in Theorem 5 with $|\Phi| = 1$, and define

$$\varepsilon_{\text{est}}(\delta') \triangleq 2B \sqrt{\frac{4 \log(8M/\delta')}{n_{\text{ev}} p_{\min}\eta}} + 2B \sqrt{\frac{2\{M \log 2 + \log(4/\delta')\}}{n_{\text{ev}}}}. \quad (61)$$

If $\hat{\tau}_{\hat{\phi}}$ is the evaluation-sample empirical-cell estimator in Proposition 5, then, with conditional probability at least $1 - \delta - \delta'$ and hence with the same unconditional probability,

$$|\hat{\tau}_{\hat{\phi}} - \tau| \leq \varepsilon_{\text{est}}(\delta') + \Gamma \left\{ \hat{D}_{\text{ev}}(\hat{\phi}) + r_{n_{\text{ev}}}^{(1)}(\delta) \right\}. \quad (62)$$

Proof. Conditionally on the training sample, the single-map version of Theorem 5 bounds $D(\hat{\phi})$ by $\hat{D}_{\text{ev}}(\hat{\phi}) + r_{n_{\text{ev}}}^{(1)}(\delta)$. The bounded-outcome cell-mean and multinomial argument gives $|\hat{\tau}_{\hat{\phi}} - \tau_{\hat{\phi}}| \leq \varepsilon_{\text{est}}(\delta')$. Theorem 2 and a union bound complete the result. $\square$

Remark 3 (Fully finite-sample enumerable regime). *For a finite representation class, the training-sample event in Theorem 5, Corollary 2, and the independent evaluation result in Theorem 7 together provide explicit representation-selection, estimation, and sensitivity errors. Their joint probability is obtained by allocating and summing the declared failure probabilities across the independent stages.*

Remark 4 (Open learning and inference components). *Three extensions remain open for the current neural implementation: an explicit neural-class deviation rate, a proved soft-to-hard approximation bound, and a joint limit theory after adaptive representation selection without honest splitting. The fixed- ϕ efficiency theorem and the enumerable-class certificate do not establish any of these extensions.*

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer four separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the finite-state experiment verify the established certificate and its failure as the proxy channel becomes ill-posed? Fourth, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation references, not implementable competitors.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate, audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem. The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

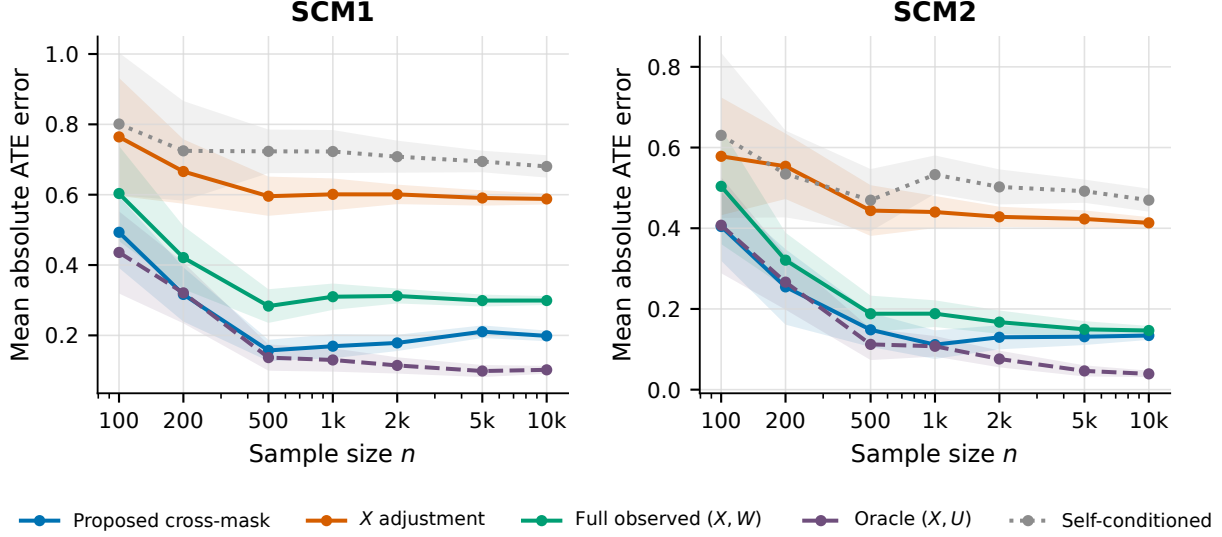


Figure 2: Honest approximate SCMs. Curves show mean absolute ATE error across twenty independently generated data sets; bands are approximate 95% Monte Carlo intervals for the mean. Prefixes are nested within a data set, so uncertainty is computed separately at each n and is not pooled across sample sizes. The proposed cross-mask learner uses one proxy coordinate for held-out auditing and averages across masks.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

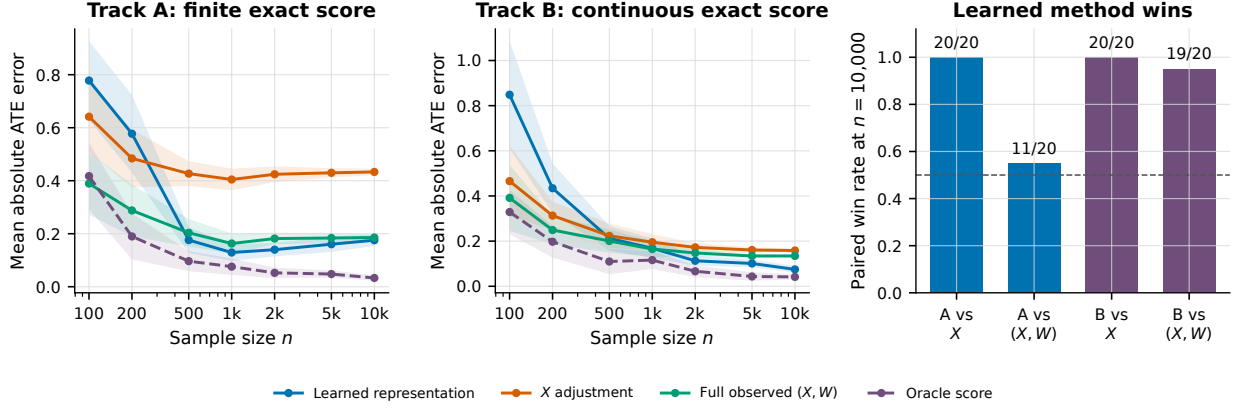


Figure 3: Exact-assumption experiments. The first two panels report mean absolute ATE error with approximate 95% Monte Carlo intervals over twenty replications. The third panel reports paired win rates at $n = 10,000$, where a win means that the learned representation has smaller absolute error than the indicated comparator. Track A satisfies the primary comparison with X but not the prespecified secondary superiority criterion against full observed adjustment. Track B records rank-recovery Spearman correlation 0.915 and held-out finite-RFF conditional-dependence surrogate 0.00691.

5.4. Finite-State Verification of the Certificate

The population verification enumerates all 3^{10} candidate maps, of which 55,980 satisfy the admissibility conditions. Exactly $3! = 6$ maps attain exact balance, and their biases are zero to machine precision. The common certificate has zero violations over all admissible maps. As the proxy channel approaches non-injectivity, the relevant stability constant grows from 1.64 to 48.9 and then diverges, even while the observable discrepancy can shrink. This is the intended sensitivity behavior: a small discrepancy is informative only together with a credible stability budget. The explicit finite-sample radius is valid but conservative in this design.

5.5. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 4. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

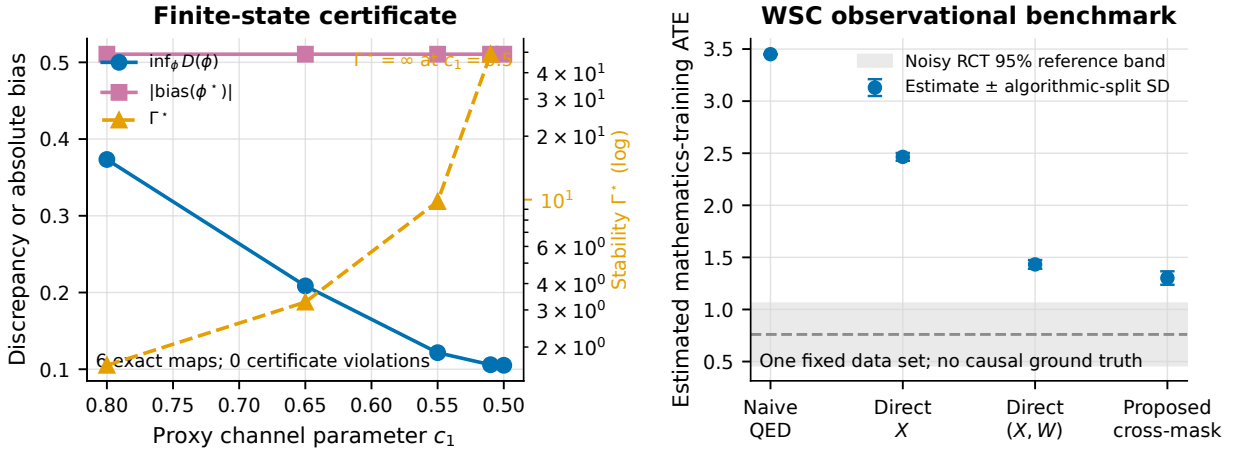


Figure 4: Certificate and observational diagnostics. Left: as the finite-state proxy channel approaches non-injectivity, the observable discrepancy decreases while realized bias remains fixed and the required stability constant diverges. Right: WSC estimates relative to the noisy randomized reference. Error bars for fitted WSC methods are standard deviations across ten algorithmic splits of one fixed data set, not Monte Carlo uncertainty. The WSC panel has no causal ground truth and reports only the unconstrained cross-mask learner as the proposed method.

6. Discussion and Limitations

6.1. What the Method Delivers

The canonical method uses a single pretreatment proxy in two roles that are available from observed data: supervising an outcome-free representation and measuring the residual imbalance of (X, W) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma D(\phi)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D(\phi) > 0$ because Γ is not generally identified from one proxy.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Proxy-channel invariance, conditional mixture injectivity, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore a testable endpoint only up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, injectivity, or stability.

6.3. Attainability and Realized Bias

For $Z = \phi(X)$ with finitely many strata, exact balance is attainable only on a narrow boundary characterized in Section 3. This is a strength as a safety certificate and a limitation as a universal identification method. In the approximate regime, reducing $D(\phi)$ reduces the certified worst-case bias under a common Γ , but it need not reduce the realized bias without additional alignment conditions. Cancellation, outcome-irrelevant proxy directions, bias floors, and ill-posed channels explain why the two orderings can differ.

6.4. Learning and Inference Gaps

The current finite-sample theorem covers enumerable representation classes and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

Cross-mask learning uses one proxy coordinate for audit and the remaining high-dimensional measurement for representation learning, then averages across audit orientations. This design blocks literal self-copying of the audited coordinate. It does not by itself establish conditional independence among proxy views, so point identification still needs a multi-view structural assumption and

an audit-channel null-space condition. The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, audit discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a proxy-supervised balancing representation, an observable discrepancy, an exact certification boundary, and a stability-indexed approximate guarantee. The paper does not claim that one generic proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks a deliberately limited question: what can an observed pretreatment proxy reveal about the hidden-confounding bias that remains after learning a low-dimensional adjustment representation? For the canonical construction $Z = \phi(X)$, exact observable joint balance identifies the ATE under the stated proxy conditions, while approximate balance yields the established sensitivity certificate $|\tau_\phi - \tau| \leq \Gamma D(\phi)$. Empirical minimization of the discrepancy selects the smallest certified worst-case bias in the prespecified class up to generalization and optimization error. The current simulations support this mechanism in exact and approximate designs and expose important failures, including self-conditioning, ill-posed proxy channels, surrogate mismatch, and practical-overlap loss. Completing the neural learning theory and independently validating the audit-view extension are the principal steps needed to turn the present manuscript skeleton into a submission-ready paper.

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A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proof Obligations

The first proof obligation is a transfer result from an observable joint balance condition to a latent balance condition. Its assumptions must state the proxy measurement relation and the injectivity or completeness property used in the transfer.

The second proof obligation is identification of τ from the observed data distribution after latent balance has been established. The proof must list every causal assumption and show where each assumption enters the identifying functional.

The third proof obligation is a boundary construction. It must exhibit two latent data-generating processes that agree on the relevant observed information but disagree on the target whenever a required transfer assumption is removed.